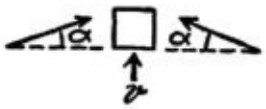
The body B moves in the positive Z direction with velocity v. The two complexes of radiation now have directions with respect to K which make an angle α with the x axis. The law of aberration states that in the first approximation α = c/v, where c is the velocity of light. From the consideration with respect to K_0 we know that the velocity v of B remains unchanged by the absorption of S and S′.



Now we apply the law of conservation of momentum with respect to the z direction to our system in the coordinate-frame K.

I. *Before the absorption* let M be the mass of B; Mv is then the expression of the momentum of B (according to classical mechanics). Each of the complexes has the energy E/2 and hence, by a well known conclusion of Maxwell's theory, it has the momentum E/2c. Rigorously speaking this is the momentum of S with respect to K_0. However, when v is small with respect to c, the momentum with respect to K is the same except for a quantity of second order of magnitude (v_2/c_2 compared to 1). The z-component of this momentum is E/2c sin α or with sufficient accuracy (except for quantities of higher order of magnitude) E/2c α or E/2c. v/c^2. S and S′ together therefore have a momentum E v/c^2 in the z direction. The total momentum of the system before absorption is therefore

$$Mv + \frac{E}{c^2} \cdot v$$

II. *After the absorption* let M′ be the mass of B. We anticipate here the possibility that the mass increased with the absorption of the energy E (this is necessary so that the final result of our consideration be consistent). The momentum of the system after absorption is then

$$M'v$$

We now assume the law of the conservation of momentum and apply it with respect to the z direction. This gives the equation

$$Mv + \frac{E}{c^2} v = M'v$$

or

$$M' - M = \frac{E}{c^2}$$

This equation expresses the law of the equivalence of energy and mass. The energy increase E is connected with the mass increase E/c^2. Since energy according to the usual definition leaves an additive constant free, we may so choose the latter that

$$E = mc^2$$